Cambridge Assessment International Education

Cambridge International AS & A Level

MATHEMATICS **9709/53**

Paper 5 Probability & Statistics 1 **May/June 2021**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mathematics Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

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Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

DM or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

FT Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

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Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

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Question	Answer	Marks	Guidance
1(a)	60	**B1**	Accept 60 or 61. No decimals
		1	
1(b)	65% of 160 = 104	**M1**	0.65 × 160 (=104) seen unsimplified or implied by use on graph
	136 (cm)	**A1**	Use of graph must be seen. **SCB1** correct value (136 only) if neither 104 nor use of graph are evident
		2	
1(c)	UQ: 150 LQ: 76 IQR = 150 – 76 = 74 [cm]	**M1**	UQ – LQ ; 148 ⩽ UQ ⩽ 152; 74 ⩽ LQ ⩽ 78.
		A1	Must be from 150 - 76
		2	

Question	Answer	Marks	Guidance
2	$p + p + 0.1 + q + q = 1$	**B1**	Sum of probabilities = 1
	$0.1 + 2q = 3(2p)$	**B1**	Use given information
	Attempt to solve two correct equations in p and q	**M1**	**Either** use of Substitution method to form a single equation in either p or q and finding values for both unknowns. **Or** use of Elimination method by writing both equations in same form (usually $ap + bq = c$) and + or – to find an equation in one unknown and finding values for both unknowns.
	$p = \frac{1}{8}$ or 0.125 and $q = \frac{13}{40}$ or 0.325	**A1**	CAO, both WWW
		4	

Question	Answer	Marks	Guidance
3(a)	Mean height $= \frac{\Sigma x + \Sigma y}{6+11} = \frac{1050+1991}{6+11} = \frac{3041}{17}$	**M1**	Use of appropriate formula with values substituted, accept unsimplified.
	178.9	**A1**	Allow 178.88, $178\frac{15}{17}$, 179
		2	

Question	Answer	Marks	Guidance
3(b)	$\frac{\Sigma x^2+\Sigma y^2}{6+11}=\frac{193700+366400}{6+11}$	**M1**	Use of appropriate formula with values substituted, accept unsimplified.
	$\text{Sd}^2=\frac{560100}{17}-\textit{their}\,178.88^2\,[=948.289]$	**M1**	Appropriate variance formula using *their* mean^2, accept unsimplified expression.
	Standard deviation = 30.8	**A1**	Accept 30.7
		3	

Question	Answer	Marks	Guidance
4(a)	[Possible cases: 1 1 2, 1 2 1, 2 1 1] Probability = $\left(\frac{1}{6}\right)^3\times 3$	**M1**	$\left(\frac{1}{6}\right)^3\times k$, where k is an integer.
		M1	Multiply a probability by 3, not +, – or ÷
	$\frac{1}{72}$	**A1**	Accept $\frac{3}{216}$ or $0.013\dot{8}$ or 0.0139
		3	
4(b)	$\text{P}(18)=\left(\frac{1}{6}\right)^3\left[=\frac{1}{216}\right]$	**B1**	
	$\text{P}(18 \text{ on 5th throw})=\left(\frac{215}{216}\right)^4\times\frac{1}{216}$	**M1**	$(1-p)^4p$, $0<\textit{their}\ p<1$
	0.00454	**A1**	
		3	

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Question	Answer	Marks	Guidance
5(a)	$z_1 = \frac{4-\mu}{\delta} = -1.378$	**B1**	$1.378 \leqslant z_1 \leq 1.379$ or $-1.379 \leqslant z_1 \leqslant -1.378$
	$z_2 = \frac{10-\mu}{\sigma} = 0.842$	**B1**	$0.841 \leqslant z_2 \leqslant 0.842$ or $-0.842 \leqslant z_2 \leqslant -0.841$
	Solve to find at least one unknown: $\frac{4-\mu}{\sigma} = -1.378$ $\frac{10-\mu}{\sigma} = 0.842$	**M1**	Use of ±standardisation formula once with μ, σ, a z-value and 4 or 10, allow continuity correction, not σ^2 or $\sqrt{\sigma}$
		M1	Use either the elimination method or the substitution method to solve two equations in μ and σ.
	$\sigma = 2.70 \quad \mu = 7.72$	**A1**	$2.70 \leqslant \sigma \leqslant 2.71 \quad 7.72 \leqslant \mu \leqslant 7.73$
		5	
5(b)	$\Phi(2) - \Phi(-2) = 2\Phi(2) - 1$	**M1**	Identifying 2 and –2 as the appropriate z-values
	$2 \times \textit{their}\, 0.9772 - 1$	**B1**	Calculating the appropriate area from stated phis of z-values which must be ± the same number
	0.9544 or 0.9545	**A1**	Accept AWRT 0.954
	$0.9544 \times 800 = 763.52$ 763 or 764	**B1 FT**	FT *their* 4SF (or better) probability, final answer must be positive integer
		4	

Question	Answer	Marks	Guidance
6(a)	$\frac{11!}{2!3!}$	**M1**	11! alone on numerator – must be a fraction. $k! \times m!$ on denominator, $k = 1, 2$, $m = 1, 3, 1$ can be implied but cannot both = 1. No additional terms
	3326400	**A1**	Exact value only
		2	
6(b)	8! = 40320	**B1**	Evaluate, exact value only
		1	
6(c)	$\frac{9!}{3!} \times 7$	**M1**	$\frac{9!}{3!} \times k$ seen, k an integer > 0, no +, – or ÷
		M1	7 × an integer seen in final answer, no +, – or ÷
	423360	**A1**	Exact value only
	Alternative method for Question 6(c)		
	${}^9C_3 \times 7! \left(\times \frac{3!}{3!}\right)$	**M1**	$9C3 \times k$ seen, k an integer > 0, no + or –
		M1	$7! \times k$ seen, , k an integer > 0, no + or –
	423360	**A1**	Exact value only but there must be evidence of $\times \frac{3!}{3!}$

Question	Answer	Marks	Guidance
6(c) cont'd	**Alternative method for Question 6(c)**		
	$3\times7\times\frac{8!}{2!}$	**M1**	$3\times\frac{8!}{2!}\times k$ seen, k an integer > 0, no + or –
		M1	7 × an integer seen in final answer, no +, – or ÷
	423360	**A1**	Exact value only
	Alternative method for Question 6(c)		
	$7\times\frac{2}{11}\times\frac{9}{10}\times\frac{8}{9}\times\frac{7}{8}\times\frac{1}{7}\times$ total no. of arrangements	**M1**	Product of correct five fractions × k seen, k an integer > 0, no + or –
		M1	7×'total no of arrangements' ×k seen, k an integer > 0, no + or –
	423360	**A1**	Exact value only
	Alternative method for Question 6(c)		
	No E between the Rs $-\frac{{}^6C_3\times3!\times7!}{3!}=100800$ 1E between the Rs $-\frac{{}^6C_2\times3!\times7!}{2!}=226800$ 2Es between the Rs $-{}^6C_1\times3!\times7!=90720$ 3Es between the Rs $-7!=5040$	**M1** **M1**	Finding the correct number of ways for no, 1 or 2 Es between the Rs, accept unsimplified. Adding the number of ways for 3 or 4 correct scenarios
	$\left[\text{Total}=7!\times(20+45+18+1)=7!\times84=\right]423360$	**A1**	CAO
		3	

Question	Answer	Marks	Guidance
6(d)	E E R _ _ $^6C_2 = 15$ E E R R _ $^6C_1 = 6$ E E E R _ $^6C_1 = 6$ E E E R R $^6C_0 = 1$	**M1**	Identifying four correct scenarios only.
		B1	Correct number of selections unsimplified for 2 or more scenario.
		M1	Adding the number of selections for 3 or 4 identified correct scenarios only, accept unsimplified. $^3C_x \times {}^2C_y \times {}^6C_z$, $x+y+z=5$ correctly identifies x Es and y Rs
	[Total =] 28	**A1**	WWW, only dependent upon 2nd M mark.
	Alternative method for Question 6(d) – Fixing EER first. No other scenarios can be present anywhere in solution.		
	E E R ^ ^ = $^8C_2$	**M1**	^8C_x seen alone or $^8C_x \times k$, , $k = 1$ or 2, $0<x<8$ Condone ^8P_x or $^8P_x \times k$, $k = 1$ or 2, $0<x<8$
		B1	$^8C_2 \times k$, $k = 1$ or 2 OE
		M1	$^8C_2 \times k$, $k = 1$ OE and no other terms
	[Total =] 28	**A1**	Value stated
		4	

Question	Answer	Marks	Guidance
7(a)(i)	$\frac{40}{800}$ or $\frac{1}{20}$ or 0.05	**B1**	
		1	
7(a)(ii)	$\frac{177}{223+177+40}$	**M1**	*Their* 223 + 177 + 40 seen as denominator of fraction in the final answer, accept unsimplified
	$\frac{177}{440}$ or 0.402	**A1**	CAO
	Alternative method for Question 7(a)(ii)		
	$P(\text{G} \mid \text{S}) = \frac{P(G \cap S)}{P(S)} = \frac{\frac{177}{800}}{\frac{223+177+40}{800}} = \frac{\frac{177}{800}}{\frac{440}{800}} = \frac{\frac{177}{800}}{\frac{11}{20}}$ or 0.55	**M1**	*Their* P(S) seen as denominator of fraction in the final answer, accept unsimplified
	$\frac{177}{440}$ or 0.402	**A1**	CAO
		2	
7(b)(i)	P(0, 1, 2) = $^{10}\text{C}_0 (0.35)^0 (0.65)^{10} + {}^{10}\text{C}_1 (0.35)^1 (0.65)^9 + {}^{10}\text{C}_2 (0.35)^2 (0.65)^8$	**M1**	One term: $^{10}\text{C}_x p^x (1-p)^{10-x}$ for $0 < x < 10$, any $0<p<1$
	0.013463 + 0.072492 + 0.17565	**A1**	Correct unsimplified expression, or better
	0.262	**A1**	
		3	

Question	Answer	Marks	Guidance
7(b)(ii)	Mean = $120\times0.35\left[=42\right]$ Variance = $120\times0.35\times0.65\left[=27.3\right]$	**B1**	Correct mean and variance seen, allow unsimplified
	P(X >32) = P($Z >$ $\frac{32.5-42}{\sqrt{27.3}}$) = P($Z$ > - 1.818)	**M1**	Substituting *their* mean and variance into ±standardisation formula (any number), condone σ^2 or $\sqrt{\sigma}$
		M1	Using continuity correction 31.5 or 32.5
	$\Phi\left(1.818\right)$	**M1**	Appropriate area Φ, from final process, must be probability
	0.966	**A1**	$0.965 \leqslant p \leqslant 0.966$
		5	